

Interpretable Machine Learning Framework for Gene Regulatory Network and Pathway Analysis in Breast Cancer

Sriporna Biswas^{1*}, Tanjidul Huda², Hema Pal³ and Dr. Vinod Kumar Gupta³

¹Department of Computer Science & Engineering, Chandigarh University, Punjab, India-140413

²Department of Biomedical Science, BCAS, University of Delhi, Delhi, India - 110007

³Rapture Biotech International (P) Ltd. D-201 Sector 10, Noida (U.P) India -201301

***Corresponding author's E-mail:** sripornab13@gmail.com

ABSTRACT

One of the significant problems in classifying stages of breast cancer is the utilization of gene expression profiles as a diagnostic and therapeutic aid. Gene expression datasets are complicated by their high dimensions, noise, and imbalance, that do not lend themselves to conventional techniques. In this work, a framework consisting of data pre-processing, biologically-relevant feature selection, and SMOTE for class balancing in conjunction with four ML models, including Logistic Regression, Support Vector Machine, Random Forest, and XGBoost, is developed and tested on binary classification of breast cancer stage based on TCGA gene expression data. Performance of four ML models is analyzed in terms of accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost achieved the best performance among the models tested, and demonstrated an adept capacity to encode complex nonlinear interactions between gene expressions. SHAP values were also used to evaluate and understand the influence of individual features on the output. Correlation analysis, gene regulatory network construction and KEGG pathway enrichment analysis were implemented to study and assess biological relevance of selected features. The proposed model provides a reliable and interpretable way to classify stages of breast cancer and can potentially serve as a clinical decision support system.

Keywords: Breast Cancer Stage Classification, Gene Expression Profiling, Machine Learning, XGBoost, SHAP (SHapley Additive exPlanations).

INTRODUCTION

Breast cancer is a leading public health issue worldwide and reliable staging is of significant importance to early diagnosis and successful therapy. Recent breakthroughs in high-throughput sequencing have provided high-dimensional gene expression profiles, revealing more insight into the molecular mechanisms of cancer and enabling the use of machine learning for the purpose of prediction[1]. There are some challenges when dealing with gene expression data: the high dimensionality of the dataset, noise of gene expression, intricate non-linear relationship among data and class imbalance, leading to the models overfitting and being unstable. Several machine learning models like LR, SVM, RF and XGBoost have been utilized for the classification of

cancer but most studies only concerned the classification accuracy and have not provided interpretable models or validated the biologically relevant features[2,3]. We developed an interpretable machine learning framework to classify breast cancer stages using TCGA gene expression data. It involved preprocessing, features selection, class balancing with SMOTE, and comprehensive assessment of multiple models[4]. We used SHAP to detect genes important to the predictions and validate biological relevance through correlation analysis, gene regulatory network modeling and KEGG pathway enrichment[5,6]. This approach allows obtaining good prediction accuracy and is highly interpretable and biologically relevant to biomedical research and clinical application.

BACKGROUND

Gene expression data have considerably contributed to the progress of cancer research by allowing the observation of changes occurring at the molecular level during disease development. Contrary to classical diagnosis techniques, which study morphological alterations, gene expression profiles analyze thousands of genes and hence offer a finer picture on the cellular processes[2,3]. Such data set the groundwork to apply machine learning algorithms for the discovery of patterns and classification problems on the biological data. Technical challenges associated with gene expression data necessitated the application of tailored machine learning techniques[4]. Gene expression datasets are typically characterized by large dimension (number of gene features) and small sample size, raising the risk of over-fitting, which reduces the generalizability of the models. They also tend to be characterized by noises, resulting from inherent biological variations and experimental context; as well as the presence of intricate, sometimes non-linear relationship among genes, which is difficult to model with traditional statistics; class imbalance, where samples associated to one cancer stage or not, is poorly represented in the training set[7,8].

Machine learning models offer appropriate solutions for the aforementioned problems. Simple models like Logistic Regression give base performance whereas Support Vector Machine is particularly adequate to address high-dimension data and non-linear decision boundaries with the application of kernels[9]. Ensemble methods like Random Forest and Gradient Boosting like XGBoost can successfully handle complex feature interactions and achieve high predictive performances[10,11]. Interpretability techniques, such as SHAP have been developed to quantify the impact of each feature on the model's prediction. Simultaneously, the methods used for biological validation, such as correlation analysis, modeling of gene regulatory networks, pathway enrichment analysis, complement machine learning based analyses and ensure the biological validity of patterns revealed[12,13]. These aspects highlight the need of building an integrated framework of machine learning, interpretability and validation for effective cancer stage classification.

LITERATURE REVIEW

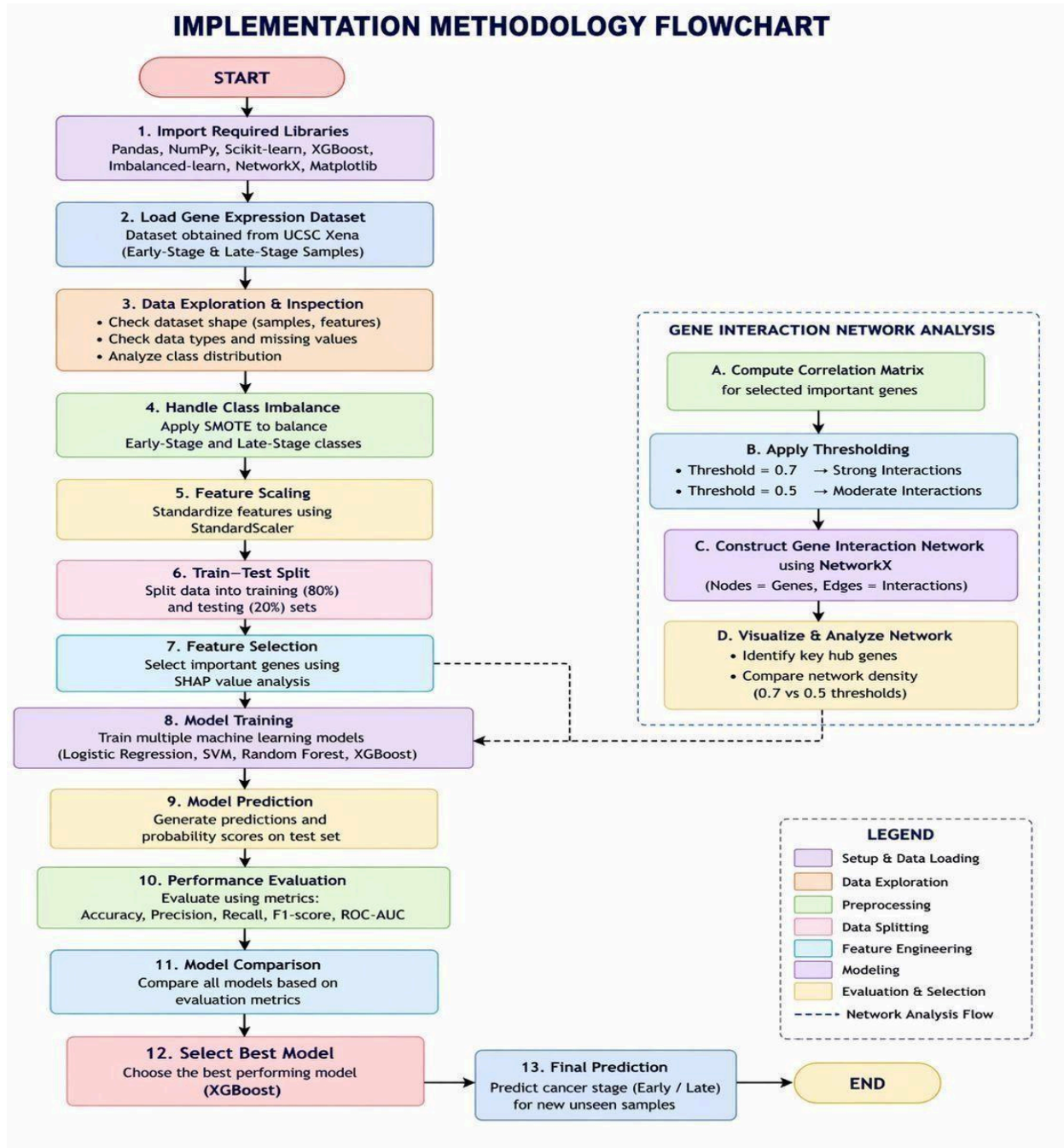
Machine learning models like Support Vector Machines, random forest, and gradient boosting models such as XGBoost have been popularly used as machine learning models used in cancer classification tasks. The task of data integration for multi-omics data has emerged as a powerful

strategy to unravel the complex biological underpinnings of cancer [1,2]. Resistance to targeted anti-cancer drugs is a complex phenomenon and a major challenge in cancer treatment [4,5]. Breast cancer subtyping plays a pivotal role in guiding personalized therapies, particularly in identifying patient-specific treatment responses and drug resistance mechanisms [6]. Female breast cancer has now exceeded lung cancer as the most diagnosed cancer, with an estimated 300,000 new cases each year (14.3%), followed by prostate (13.6%), lung and bronchus (11.28%), and colorectal (7.24%) cancers (Siegel et al. 2023) [7]. Mining gene regulatory network (GRN) is an important avenue for addressing cancer mechanisms [14,15]. Differential gene expression and LASSO (Least Absolute Shrinkage and Selection Operator) regression are employed for reducing the omics data dimensionality and feature selection; hence, the developed models are referred to as LASSO-MOGCN, LASSO-MOGAT, and LASSO-MOGTN. Graph structures constructed using sample correlation matrices and protein-protein interaction networks are investigated [16]. There are several types of breast cancer based on the origin and behavior of the tumor cells. These include Lobular Carcinoma in Situ (LCIS), Ductal Carcinoma in Situ (DCIS), Infiltrating Lobular Carcinoma (ILC), Infiltrating Ductal Carcinoma (IDC), Inflammatory Breast Cancer (IBC), and Infiltrating Mucosecretor Carcinoma (IMC). Among them Infiltrating Mucosecretor Carcinoma (IMC) also known as mucinous carcinoma, is a rare subtype of breast cancer that accounts for less than 5% of all cases of invasive breast cancer [17]. In the case of cancer complexities, dysregulated pathways including DNA damages leading to Epithelial-Mesenchymal Transition (EMT), cell proliferation, morphogenesis, as well as dissemination of tumor cells can emerge during different breast cancer stages [18,19]. We further evaluate methods on simulated datasets to probe sensitivity with controlled signal and noise [20]. Future directions will incorporate prospective, multi-centric validation; imaging and radiomics; enhancement to site differences and missing data; tracking of model calibration in real-life use.[21,22]. Though gene expression and machine learning based cancer classification have obtained great success, still there are many limitations for these works in the present body of knowledge [23]. A large portion of the methods focuses only on improving the accuracy and lacks the interpretability, so it is not easy to understand the reason for prediction. The issues of class imbalance in biomedical data most commonly do not get enough attention leading to biased predictions[24]. Also biological validation methodologies are missing from most of the works and thus the identified features may not be biologically relevant to the disease mechanism [25]. There is a need for the future an integrated framework of high performance machine learning methods and interpretability with biological validation for an effective classification of cancer stage.

METHODOLOGY

Following the outlined structure, a complete machine learning framework for the classification of breast cancer stage is developed. The problem, the task of distinguishing between early-stage and late-stage breast cancer on the basis of gene expression levels is viewed as a supervised binary classification problem. Since the available raw dataset includes a high-dimensional gene expression profile and the number of samples are far fewer than the number of genes, the risk of overfitting is high due to noise, variance, etc.

The overall workflow of the proposed system is illustrated in Figure 1.



Initial preprocessing is performed by normalization and data cleaning to ensure the uniformity of gene expression levels of various samples. Then, feature selection is implemented to overcome the high-dimensional problem and remove irrelevant or redundant genes. Instead of using all genes in the feature space, 70 genes known to be associated with cancer-related pathways are chosen to increase efficiency and minimize the loss of biological information, and also to

overcome overfitting, to prevent noise, and to achieve good generalization. In addition, class imbalance is observed between early-stage and late-stage cancer patients, thus, oversampling using SMOTE is applied to increase the number of samples of minority classes to improve the fairness of the classifier. Several machine learning algorithms including Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost are developed to perform classification tasks, Logistic Regression is taken as a baseline due to its simple structure; SVM can handle high-dimensional data; Random Forest builds multiple decision trees and uses ensemble learning strategy; XGBoost is a gradient boosting framework which can outperform the other algorithm. The dataset is divided into training and testing sets, and cross-validation is applied on the training set to prevent overfitting. Finally, multiple evaluation metrics including accuracy, precision, recall, F1-score, and ROC-AUC are calculated to assess the performance of the classifiers and to compare different algorithms, which allows us to select the optimal algorithm for our breast cancer classification. As the machine learning model usually is a black-box model, we perform SHAP to understand how the selected features impact on classification; then, biological validations were carried out to evaluate the biological significance of identified genes, correlation analysis is implemented to find the relation between genes, and Gene regulatory network modeling, hub gene identification, and KEGG pathway enrichment analysis are carried out to evaluate the biological significance of selected genes and their involvement in cancer-associated pathways.

RESULTS AND DISCUSSION

Experimental Setup

Our experiment on gene expression data was in combination with TCGA data prepared by the UCSC Xena platform. The original number of samples was 1219, among which 900 samples were of early stages and 300 samples of late stages. To minimize the effect of an imbalanced data distribution between early and late stages, SMOTE (Synthetic minority oversampling technique) was used and amounted to 1652 samples after balancing. A subset of biologically relevant genes was chosen and the datasets were normalized with feature scaling. The dataset was prepared as a training and testing set and different machine learning algorithms were trained and tested with various performance measurements. Experimental Setup

The experiment was performed on gene expression data that the UCSC Xena retrieved from the TCGA database. At the start there were 1219 samples, with 900 early stage and 300 late stage primary breast cancers. SMOTE was used to oversample the data set. The outcome was a total of 1652 samples, with an equal number of early and late stage cases. A selection of biologically relevant genes were chosen; the dataset was then normalized and scaled for all features, through feature scaling. The data was then split into training and testing sets, using many machine learning models and reporting the resulting performance assessments. From Table 1, the characteristics of the dataset show the clear class imbalance of the original data, which contains a much larger number of early stage samples than late stage samples. Applying SMOTE technique balances the dataset with identical proportions of early stage and late stage. Balancing class is critical to decrease the bias of the models and achieve better classification performance. The further pre-processing methods (feature selection and normalization) are used to improve the

data quality. This will provide better performance for machine learning models to extract useful knowledge.

Table 1: Dataset Characteristics and Class Balancing.

Category		Dataset Characteristics	Details
0	Source	Dataset Source	UCSC Xena
1	Original	Total Samples (Original)	1219
2	Original	Early Stage (Original)	900
3	Original	Late Stage (Original)	300
4	Balanced	Total Samples (After SMOTE)	1652
5	Balanced	Early Stage (Balanced)	826
6	Balanced	Late Stage (Balanced)	826

Model Performance

All four classification models (Logistic Regression, Support Vector Machine (SVM), Random Forest and XGBoost) were trained and tested on the data. There was similarity across all the four models with regards to classification accuracy with all reaching roughly 83% accuracy in their classification. Where these models differed was in other metrics which provide a more detailed understanding of their performance. These include recall, precision and ROC-AUC. In relation to the ROC-AUC, it is noticeable that XGBoost had the highest ROC-AUC of 0.626 and Logistic Regression followed with an ROC-AUC of 0.615. Both SVM and Random Forest had a relatively poorer performance at 0.562 and 0.543 respectively. Although recall was good for all four models, XGBoost struck the best balance between precision and recall values.

Table 2: Model Performance Comparison.

===== TABLE 2: MODEL PERFORMANCE =====						
	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
3	XGBoost	0.832	0.832	1.000	0.908	0.626
0	Logistic Regression	0.832	0.835	0.995	0.908	0.616
1	SVM	0.832	0.832	1.000	0.908	0.562
2	Random Forest	0.832	0.832	1.000	0.908	0.543

Model Comparison:

From the comparison, all models got the same accuracy. However, XGBoost demonstrated better performance over other models with ROC-AUC. It was demonstrated that the XGBoost model was able to catch the complicated non-linear relationships in high-dimensional gene expression data. Gradient boosting framework in XGBoost will perform to minimize the prediction error one by one, thereby improving the classification result compared to models like LR and SVM.

Though SVM and RF got the recall accuracy equal to 1, the value of ROC-AUC was smaller than XGBoost which demonstrates the boundary of the classification model was not as strong as XGBoost. LR worked fine but it is based on linear relationships and is not ideal for the complex data. Thus, the XGBoost model was chosen for this study.

Receiver Operating Characteristic (ROC) Analysis:

The results of trained models were also analyzed by the ROC curves. A Receiver Operating Characteristic (ROC) curve is a plot which illustrates the performance of a binary classification system as the classification threshold is varied.

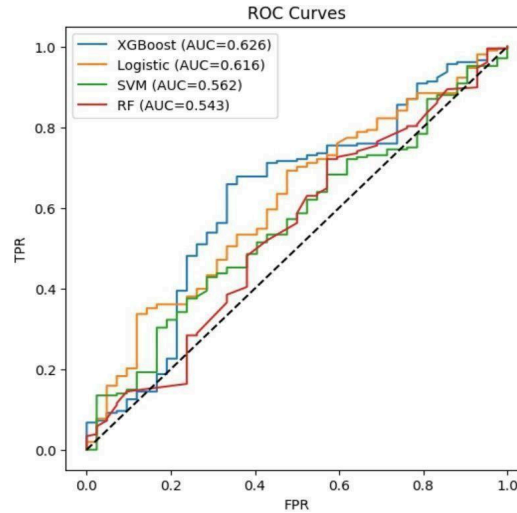


Figure 2: ROC curve comparison of machine learning models showing that XGBoost achieves the highest AUC, indicating better classification performance.

The classification performance can be comprehensively assessed by drawing the ROC curve (as shown in Figure 2) where the tradeoff between TPR and FPR at various thresholds is clearly visible. The larger the value of AUC (Area Under the Curve), the better it is at distinguishing samples at early stages from those at later stages of the cancer. The XGBoost classifier achieves the largest value among all the classifiers presented here. In Figure 2, it is clearly shown that the ROC curve for XGBoost is consistently higher than for Logistic Regression, SVM and Random Forest and hence has the highest discriminative ability with different thresholds, and although recall for SVM and Random Forest is high their ROC curve closer to the diagonal represents poor discriminative power across thresholds whereas Logistic regression has good value and discriminative power, however, they are not able to model the complex non-linear relationships present in gene expression data. XGBoost models the errors from weak prediction rules through gradient boosting and is effective at finding the complicated non-linear structure within high-dimensional data hence maintains a good tradeoff between sensitivity and specificity hence can serve as a good classifier for classifying cancer stages. Thus, for this study it is observed that XGBoost is the most effective model based on ROC-AUC.

Model Interpretability using SHAP:

To increase the interpretability, SHAP analysis on the XGBoost was performed to detect the importance of genes in the classification. From the SHAP summary plot it shows that genes

RUNX1, ERBB2, NCOR2 and FOS are effective on the classification model, and suggest that these genes are significant to differentiate the cancer stages.

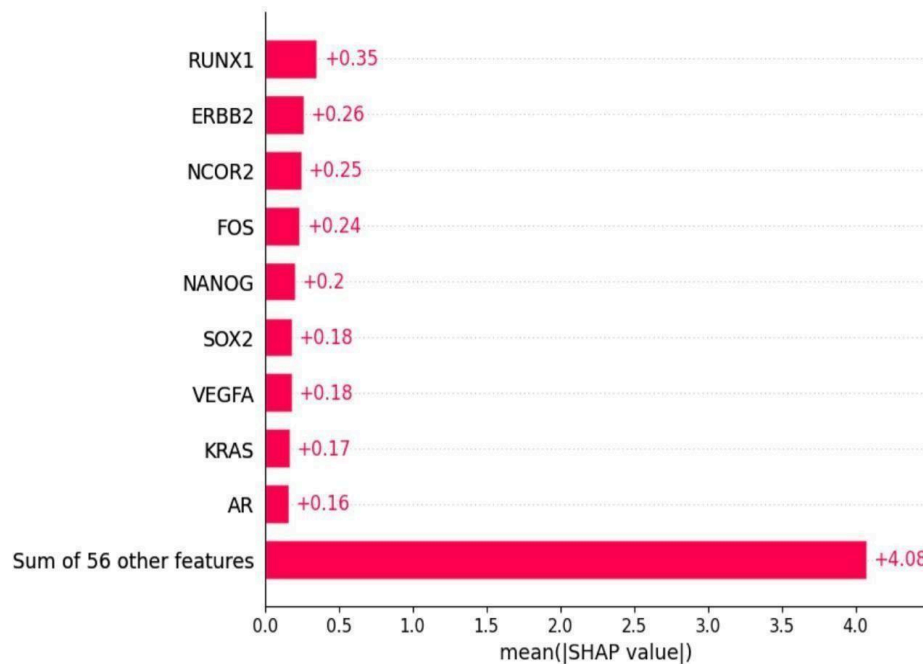


Figure 3: SHAP summary plot showing the contribution of top genes to model predictions.

The SHAP value method provided a quantitative interpretation of feature importance by demonstrating each gene's impact on the classification outcome. The higher the SHAP value for a gene, the greater its contribution towards classifying a sample as early-stage or late-stage cancer. It was observed that RUNX1, ERBB2, NCOR2 and FOS were the features that contributed most to the model's classification outcome.

In addition to indicating feature importance, SHAP values illustrate the direction of each feature's impact – i.e. Whether increased or decreased levels positively or negatively affect the outcome. This makes for a richer insight into the contribution each individual gene expression feature contributes to the model behavior, making it far less of a black box than standard classification models by associating the classification outcome with specific biological features. The biologically significant genes with the highest SHAP scores and whose gene expression patterns strongly correlate with disease progression add further validity to the classification model, thus producing a high performance and biologically significant model.

Biological Validation:

In order to be sure that selected features are not only statistically significant but also biologically meaningful, a robust strategy of biological validation was performed. Gene expression datasets are high-dimensional and complex data, and the model may reveal biological patterns which have no relevance. Therefore, the validation that selected genes are functionally related to breast cancer progression is necessary. Multiple validation strategies, including analysis of gene expression patterns, correlation analysis, construction of gene interaction networks, identification

of hub genes, were applied in this study. Through analysis from different views, deeper insights were revealed of how genes function individually, how genes are regulated and co-regulated and in which mechanism they play a role in breast cancer progression.

Interpretation of Gene Expression Heatmap

The heatmap of gene expression presents a holistic view of the difference in expression of the selected genes in all the samples. A row corresponds to a single sample and a column to a particular gene. The identified difference in the intensity of color among various samples indicates unique gene expression patterns at various stages of breast cancer.

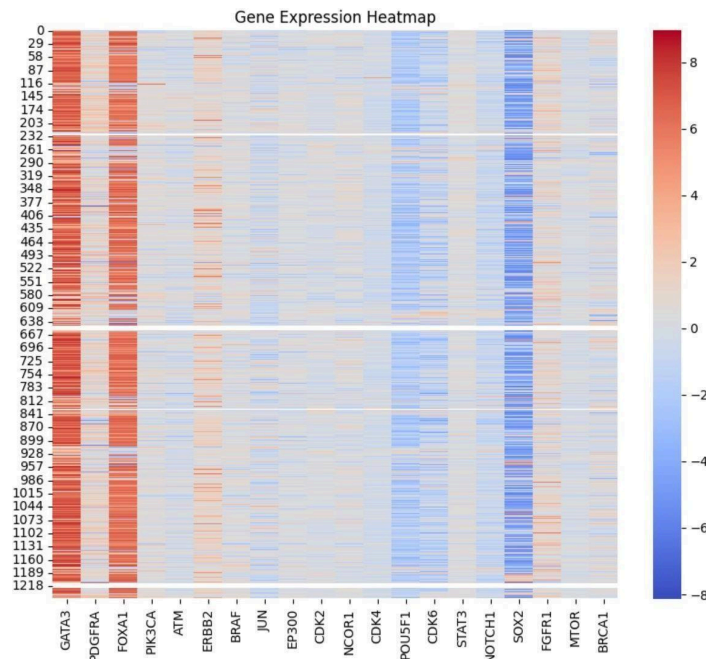


Figure 4: Gene expression heatmap illustrating differential expression patterns across early and late-stage breast cancer samples.

The heatmap shows that there are notable distinctions in gene expression patterns between early-stage and late-stage samples, indicating that there are stage-specific molecular signatures. Color intensity codes the level of expression of a gene, with red being a strong expression level and blue being a weak expression level, and intermediate levels of the color indicate an intermediate level of expression of the gene. The difference in the color distribution among samples indicates the heterogeneity of gene expression in breast cancer. The genes that show uniform red patterns in many samples indicate that there is sustained overexpression and is likely to play crucial roles in tumor evolution. Conversely, the genes with dominant blue patterns show a decreased expression, which might be linked to suppressed biological processes or pathways. Notably, various genes show similar changes in their expression levels between early and late phases, indicating that they have the potential to be used as biomarkers to classify the cancer stage. Also, groups of genes with similar expression patterns may indicate that they are co-regulated or share common biological processes. The heatmap is a highly expressive

exploratory tool to identify key genes and expression patterns in high-dimensional data, both to support biological interpretation and to support selection of features to be used in machine learning-based classification models.

Interpretation of Gene Correlation Heatmap

The gene correlation heatmap has a quantitative representation of correlations between the genes of interest by quantifying how strong and directional their pairwise correlations are. The value of each cell of the matrix is the correlation coefficient between two genes, the extent to which they all vary together across samples.

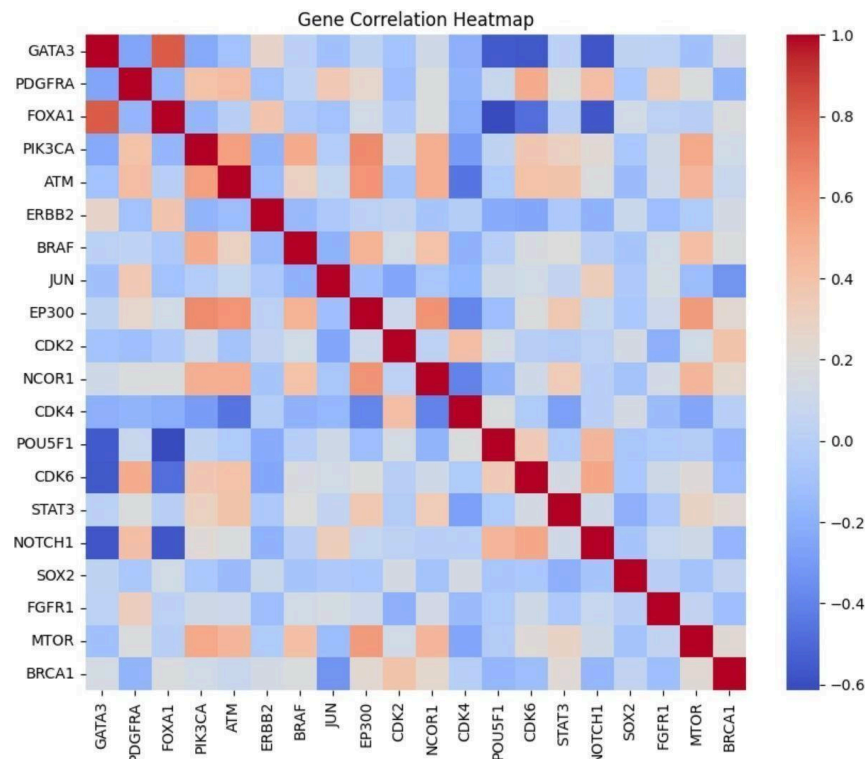


Figure 5: Gene correlation heatmap showing pairwise relationships among selected genes.

The intensity of colors in the heatmap is used to encode the degree and direction of correlation where red indicates strong positive correlation (genes increase or decrease together), blue indicates negative correlation (genes vary inversely) and light colors indicate weak or no correlation. This is because the occurrence of strongly correlated pairs of genes implies a coordinated biological process, possibly involving co-regulation or being engaged in common biological pathways. As an example, groups of positively correlated genes may be clusters of genes involved in a common set of signaling pathways or functional processes involved in cancer progression. On the other hand, pairs of genes that are negatively correlated could indicate a relationship of regulation or inhibition whereby the activation of one gene is linked to the suppression of another gene. These patterns give clues regarding the possible hostile biological processes. The consistency of the correlation matrix is verified by the symmetry of the

heatmap and the high values of the diagonal (self-correlation = 1). In addition to individual relationships, the overall pattern shows groups of genes as correlation clusters, and thus highlighting potential functional modules within the gene network. In general, the correlation heatmap is a critical analytical tool that can be used to identify key regulatory interactions and support the analysis of the downstream (e.g., building a gene network and identifying hub genes).

Gene interaction network construction

We calculated gene interaction networks based on different correlation thresholds to quantify the relationships between genes. A gene interaction network generated by higher threshold (0.7) displayed strongly correlated gene pairs, which can reveal more crucial connections among influential genes. Then the correlation threshold was decreased to 0.5 to produce a more densely connected network of genes with less strongly correlated.

From increasing interconnection between genes in the network, it can be seen that there are only a few genes that have direct strong relations to each other, but a set of genes have mutual relationships to jointly regulate the system. The dense network with a number of interconnected nodes also indicates the complicated regulatory process in breast cancer development.

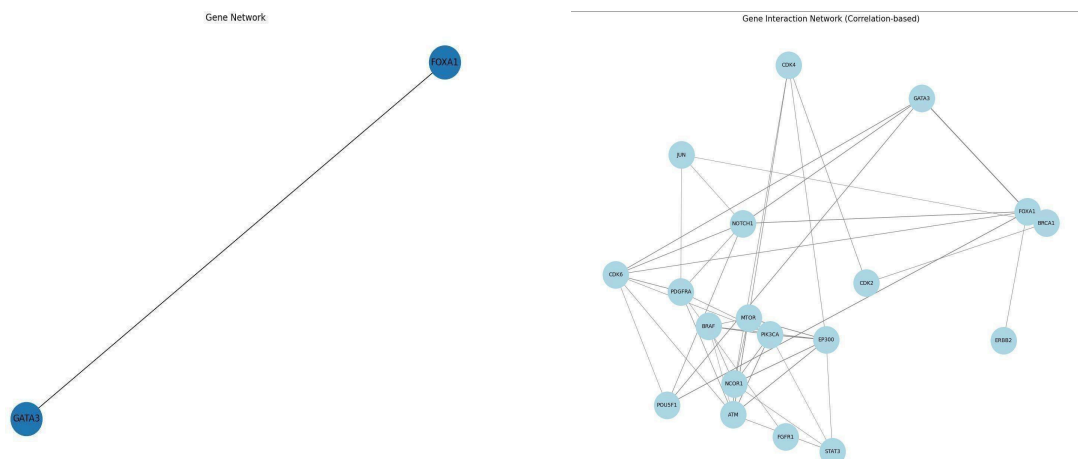


Figure 6: gene interaction network calculated based on different correlation thresholds. Hub Gene Identification and Network Centrality Analysis.

Connectivity-based network analysis was carried out to identify most influential genes in the gene interaction network. In this method, the genes are modeled as nodes and their interaction (based on correlation relationship above a given threshold) as edges. The significance of each gene in the network was measured using degree centrality which is the number of direct links that a given gene has with the rest of the genes in the network.

Genes that have a higher connectivity (so-called hub genes) are biologically important because they tend to regulate many pathways and can be central to maintaining network stability and functionality. The genes that showed the highest level of connectivity in the current study are

ATM, PIK3CA, CDK6 and NCOR1, which means that they take the leading role in the interaction network.

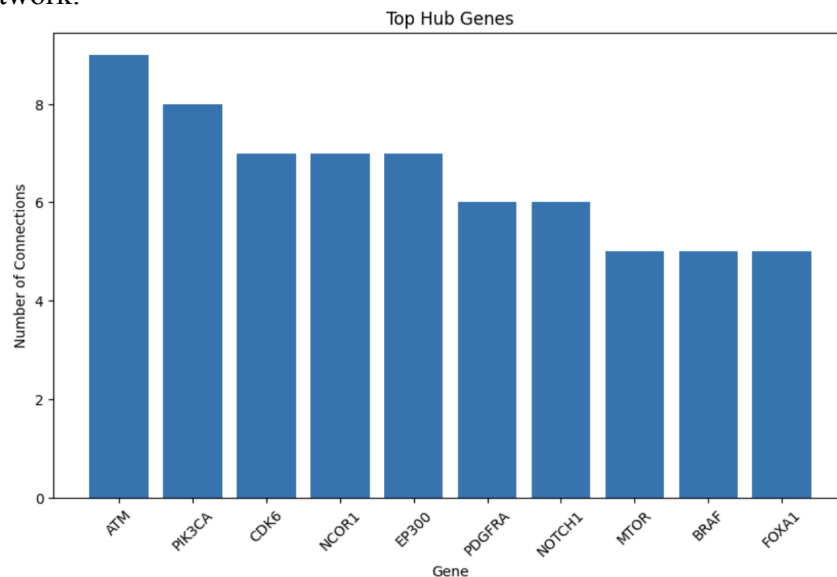


Figure 7: Top hub genes identified based on network connectivity.

The number of connections to each gene is shown in the bar graph, which is a quantitative comparison of the relative importance of the two genes. The genes with higher bars are associated with larger interaction density, which implies that they become key regulators or control points of the biological system. The discovery of hub genes is important in a number of ways. To start with, hub genes play a role in the many important processes in cells including cell cycle regulation, signal transduction and apoptosis which are key processes in the development and progression of cancer. Secondly, since they are located centrally within various pathways, they can be considered as potential biomarkers to be utilized in the diagnosis or prognosis of a disease. Thirdly, hub genes represent very promising therapeutic targets, because the activity of these hubs can have global effects on the entire gene network. Moreover, the high connectivity between genes adds to the biological validity of the constructed network, which implies that the observed interactions are not accidental but represent the effective regulatory relationships. The similarity of the identification of hub genes and the patterns of previously known gene expression and correlation lends the model greater reliability. On the whole, this discussion shows that only a small number of highly connected genes contribute a disproportionately large portion to the progression of breast cancer. These hub genes are not only important in increasing the interpretability of the model but also offer useful insights into the underlying molecular mechanisms, which makes them critical candidates to further biological and clinical studies.

KEGG Pathway Enrichment Analysis

KEGG pathway enrichment analysis was carried out to further explore the biological significance of the identified genes. Pathway enrichment analysis is able to identify if the chosen genes are significantly enriched in already known biological pathways linked to the progression of the cancer and the development of the tumor.

The analysis showed the attributes of several cancer related pathways were enriched among the selected genes, which showed that they have a strong association with breast cancer biology.

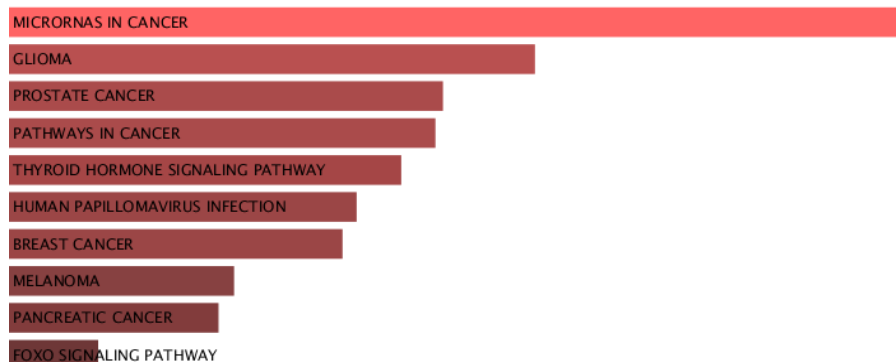


Figure 8: KEGG pathway enrichment analysis showing significantly enriched cancer-related pathways associated with selected genes.

The pathway enrichment results show that the genes identified are involved in several oncogenic and regulatory pathways. The pathway with the highest enrichment score was “MicroRNAs in Cancer”, indicating that the regulation of genes by post-transcriptional control of microRNAs is a key factor in breast cancer progression. In addition, microRNAs are known to control gene expression by binding to messenger RNA and thus affecting tumor growth, metastasis and drug resistance, as well as tumor apoptosis.

The “Glioma” and “Prostate Cancer” pathways were also highly enriched, suggesting that some of the genes in this study might have common molecular mechanisms across different cancer types. This indicates that many signaling pathways are shared and that regulatory genes that control cell growth and survival are conserved. The enrichment of the general “Pathways in Cancer” category further corroborates the very close association of the selected genes with fundamental cancer mechanisms related to uncontrolled cell growth, angiogenesis, invasion and metastasis. Further, the enrichment of genes in the pathway “Breast Cancer” also confirms the biological relevance of the selected genes to breast cancer progression and stage classification.

Cell cycle control, apoptosis, response to oxidative stress, and cellular metabolism are controlled by important regulatory pathways including the “Thyroid Hormone Signaling Pathway” and “FOXO Signaling Pathway”. These pathways have been previously reported to be dysregulated in tumor development and progression. They are enriched, implying they may play a role in molecular changes seen in breast cancer samples that are associated with abnormal signaling activity.

The enrichment of “Human Papillomavirus Infection” related pathways might be related to the activation of viral-response and immune-related signaling mechanisms that are shared with cancer-associated pathways. Likewise, pathways linked to melanoma and pancreatic cancer show common oncogenic signaling pathways between various cancers.

In summary, the KEGG pathway enrichment analysis is an excellent method of biological validation of the machine learning approach since the genes obtained from the approach were shown to be functionally associated with important cancer pathways. These findings confirm the reliability of the biomarkers selected, and provide additional confidence on the interpretability of the proposed model for breast cancer classification.

CONCLUSION

This research is an interpretable machine learning model to classify stages of breast cancer using gene expression data. The proposed pipeline combines preprocessing, feature selection, model balancing (SMOTE), model training, model evaluation and model interpretability into a single system.

XGBoost proved to be the best of all the implemented models due to the fact that it is able to capture complex non-linear relationships in high-dimensional data. SHAP uses increased model transparency by determining the key genes to affect classification, transforming the system into an explainable AI system. Additional validation by heatmap analysis, correlation of genes and network-based identification of hub genes showed some useful biological trends. The strong regulatory role was seen in key genes that included ATM, PIK3CA, CDK6 and NCOR1 and KEGG pathway enrichment analysis verified that the identified genes are significantly involved in multiple cancer-related signaling pathways, including breast cancer, microRNA regulation, and FOXO signaling pathways. The framework is a valid instrument in classifying cancer stage and analyzing the genes at the gene level.

Although the suggested framework shows good performance and analysis power, there are some opportunities to enhance and expand the framework. The combination of multi-omics data, such as proteomics, metabolomics, and epigenomics is one of the overall directions of future work. Integrating several sources of data can give a more in-depth view on biological systems and increase the accuracy of models by including a variety of interactions on a molecular level. Deep learning approaches, including Graph Neural Networks (GNNs) and deep neural architectures, are another important extension that is capable of better modeling complex gene interactions and network structures. Externally testing with independent data is also a very important step towards enhancing the strength and external validity of the model. It is also possible to direct future work at real-time clinical integration whereby the developed system can be integrated into decision-support tools for healthcare professionals. This would entail optimization to enable scalability, computational efficiency and user friendly interfaces. The study can be enhanced by expanding pathway analysis and including drug-target interaction studies that can be used to translate computational findings to therapeutic applications to enable personalized medicine and targeted treatment strategies.

Conflict of interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

REFERENCES

1. Alharbi F, Vakanski A, Zhang B, Elbashir MK, Mohammed M. Comparative analysis of multi-omics integration using graph neural networks for cancer classification. *IEEE Access*. 2025;13:37724-37736.
2. Zohari P, Haghiri Chehreghani M. Graph neural networks in multi-omics cancer research: a structured survey. *arXiv*. 2025;arXiv:2506.17234.
3. Aghaeifar R. A multifaceted approach using discrete-time survival analysis, network science, and graph neural networks to predict relapse in breast cancer. *ProQuest Diss Theses*. 2025;32237294.
4. Li AZ, Du Y, Liu Y, Chen L, Liu R. Benchmarking computational methods for multi-omics biomarker discovery in cancer. *Brief Bioinform*. 2025;bbae200.
5. Kourou K, Exarchos TP, Exarchos KP, Karamouzis MV, Fotiadis DI. Machine learning applications in cancer prognosis and prediction. *Comput Struct Biotechnol J*. 2015;13:8-17.
6. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, Mesirov JP. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102(43):15545-15550.
7. Patel JC, Shakyawar SK, Sethi S, Guda C. GAIN-BRCA: a graph-based AI-net framework for breast cancer subtype classification using multiomics data. *Bioinform Adv*. 2025;5(1):vbaf116.
8. Thakur T, Batra I, Luthra M, Vimal S, Dhiman G, Malik A, et al. Gene expression-assisted cancer prediction techniques. *J Healthc Eng*. 2021;2021:4242646.
9. Tabassum N, Kamal MAS, Akhand MAH, Yamada K. Cancer classification using high-dimensional gene expression data. *BioMedInformatics*. 2024;4(2):1275-1288.
10. Akter S, Adesola RO, Basnet S, et al. Machine learning approach to identify significant genes and classify cancer types from RNA-seq data. *Glob Med Genet*. 2025;12(4):100079.
11. Osama S, Elhoseny M, Elngar AA, et al. Gene reduction using feature selection and machine learning algorithms for cancer diagnosis. *Expert Syst Appl*. 2023;213:119270.
12. Gupta S, Sharma R, Kumar P, et al. Deep learning techniques for cancer classification using gene expression data. *Front Physiol*. 2022;13:952709.
13. Al-Azani S, Alreshidi M, Aljame M, et al. Gene expression classification with imbalance handling: a comprehensive review. *Biomed Res Int*. 2024;2024:10889442.
14. Mukhopadhyay D, Chatterjee S, Ghosh S, et al. Machine learning classification using high-dimensional gene data. *Preprints*. 2024;2024012067.
15. Biswas S, Khastagir M, Hasan MM, Reshma. From brain to machine: leveraging neuromorphic computing for improved system efficiency. *AIP Conf Proc*. 2025;3343(1):030012.
16. Zhang X, Li Y, Wang H, et al. Identifying biomarkers for breast cancer by gene regulatory network rewiring. *BMC Bioinformatics*. 2021;22:425.

17. Zhang X, Li Y, Wang H, et al. Re-wiring and gene expression changes of AC025034.1 and ATP2B1 play complex roles in early-to-late breast cancer progression. *BMC Genomics*. 2021;22:863.
18. Kanehisa M, Furumichi M, Sato Y, Kawashima M, Ishiguro-Watanabe M. KEGG: integrating viruses and cellular organisms. *Nucleic Acids Res*. 2023;51(D1):D587-D592.
19. Li R. Machine learning based cancer classification using gene expression data. *Int J Mach Learn*. 2024;14(2):48-53.
20. KalaiSelvi B, et al. Gene expression profiling with machine learning and biosensors for cancer biomarker discovery. *Sci Rep*. 2025;15:30366.
21. Kantha D, Loew MH. Multimodal fusion of imaging and multi-omics data for enhanced breast cancer detection. In: *Applied Imagery Pattern Recognition*. Cham: Springer Nature Switzerland; 2026. p. 522-545.
22. Waqas A. From graph theory for robust deep networks to graph learning for multimodal cancer analysis [dissertation]. Tampa (FL): University of South Florida; 2024.
23. Seminario Yrigoyen CA. Investigating LIONESS-derived gene regulatory networks and advanced graph neural networks for leukemia subtype classification and cross-cancer disease association analysis [master's thesis]. Turin (Italy): Politecnico di Torino; 2026.
24. Li Y, et al. Integrated network pharmacology, machine learning and experimental validation to identify the key targets and compounds of TiaoShenGongJian for the treatment of breast cancer. *Onco Targets Ther*. 2025.
25. Jin T, Nguyen ND, Talos F, Wang D. ECMarker: interpretable machine learning model identifies gene expression biomarkers predicting clinical outcomes and reveals molecular mechanisms of human disease in early stages. *Bioinformatics*. 2021;37(8):1115-1124.